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Pawl disc, lace tightening and loosening device, wearable clothing, and storage device

Abstract

A lace tightening and loosening device, wearable clothing, and a storage device are provided. The lace tightening and loosening device includes a base, a spool, a pawl disc, a dial and a connecting structure. When the dial is rotated in a first direction, the driving member drives the pawl disc to rotate in the first direction and drive the connecting structure to rotate, thereby driving the spool to rotate in the first direction to tighten a lace; when the dial is rotated in a second direction opposite to the first direction, the driving member drives the pawl to be separated from the first meshing part, so that the pawl disc rotates in the second direction and drives the connection structure to rotate.

Inventors:	**Xiang; Feng (Guangdong, CN)**
Applicant:	**SHENZHEN ETIE TECHNOLOGY LTD.** (Guangdong, CN)
Family ID:	**1000008544906**
Assignee:	**SHENZHEN ETIE TECHNOLOGY LTD. (Shenzhen, CN)**
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Primary Examiner: Upchurch; David M

Attorney, Agent or Firm: HOWARD M COHN and Associates, LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2025/081474, filed on Mar. 8, 2025, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD
(1) The present invention relates to a pawl disc, a lace tightening and loosening device, wearable clothing, and a storage device.
BACKGROUND
(2) In order to make some articles for daily use suitable for different users, many articles for daily use are designed with draw laces, such as sneakers, hats, sports shirts, and gloves. The tightness of an article is adjusted by knotting the draw lace, but in daily life, the knotted draw lace is sometimes easily loosened and untied. If shoelaces are loosened, the shoelaces easily get dirty and trip over a user, and the like, especially for young children who cannot tie laces by oneself. It is very inconvenient for the elderly to bend over to tie laces because of poor health, which easily causes danger.
(3) A lace loosening device in the current market can tighten and loosen a lace. For example, when the lace loosening device is used on a shoe, it is convenient for a user to put on and take off the shoe. However, when this lace loosening device loosens a lace, the lace needs to be completely loosened. As a result, during use, when a user feels uncomfortable if a lace is excessively tightened, the user has to completely loosen the lace and then tighten the lace to an appropriate extent. The user experience is poor.
SUMMARY
(4) In order to solve the technical problem, the present invention provides a lace tightening and

loosening device, wearable clothing, and a storage device.

(5) In a first aspect, the present invention provides a lace tightening and loosening device, which includes a housing, a spool, a pawl disc, a dial and a connecting structure.

(6) The housing includes a barrel-shaped main body, a first meshing part arranged on an inner side of the barrel-shaped main body, a partition plate arranged on the inner side of the barrel-shaped main body, and a first threading hole penetrating through the barrel-shaped main body.

(7) The spool is arranged inside the barrel-shaped main body and includes a first base plate, a second base plate, and a side wall connecting the first base plate to the second base plate. The first base plate is opposite to the second base plate; the first base plate resists against the partition plate, and the second base plate is located on one side of the first base plate away from the partition plate; the first base plate, the second base plate, and the side wall form a winding region for storing a lace; the side wall is provided with a second threading hole that penetrates through the side wall.

(8) The pawl disc is arranged on one side of the first base plate away from the second base plate and includes a substrate with a first through hole, a pawl connected to the substrate, and a first clamping part arranged on the substrate. The pawl is configured to be meshed with the first meshing part.

(9) The dial is arranged on one side of the pawl disc away from the spool and includes a cover plate, a side plate connected to one side of the cover plate, a driving member arranged on one side of the cover plate facing the side plate, and a second clamping part arranged on one side of the cover plate facing the side plate. The cover plate and the side plate are enclosed to form an accommodating cavity with an opening; a portion of the barrel-shaped main body is located in the accommodating cavity; the second clamping part is clamped with the first clamping part.

(10) The connecting structure is arranged in the first through hole in a penetrating manner and connecting the pawl disc to the spool. The connecting structure is movably connected to the spool.

(11) When the dial is rotated in a first direction, the driving member drives the pawl disc to rotate in the first direction and drive the connecting structure to rotate, thereby driving the spool to rotate in the first direction to tighten a lace; when the dial is rotated in a second direction opposite to the first direction, the driving member drives the pawl to be separated from the first meshing part, so that the pawl disc rotates in the second direction and drives the connection structure to rotate; meanwhile, under an external force, the lace drives the spool to rotate in the second direction and be loosened from the spool; when no external force is applied, the spool stops rotating; when the dial is lifted axially away from the housing under an external force, the pawl disc and the connecting structure are away from the spool, so that the lace is completely loosened from the spool under the external force.

(12) Preferably, the connecting structure includes a connecting piece arranged in the first through hole; the connecting piece includes a main body part, a connecting block arranged at the main body part, and a first tooth arranged at one end of the main body part; the pawl disc further includes a connecting slot extending in an axial direction of the substrate; the connecting slot is communicated to the first through hole; the connecting block is arranged in the connecting slot and slides along the connecting slot; a second tooth is arranged on one side of the first base plate away from the second base plate; and the second tooth is configured to be meshed with the first tooth, so that the connecting piece pushes the spool to rotate in the first direction, and the connecting piece rotates in the second direction to be demeshed from the spool.

(13) Preferably, the first tooth includes a first inclined surface and a first arc-shaped surface facing away from the first inclined surface; the second tooth includes a second inclined surface and a second arc-shaped surface facing away from the second inclined surface; the first inclined surface is configured to resist against the second inclined surface, which enables the connecting piece to push the spool to rotate in the first direction; and the first arc-shaped surface is configured to resist against the second arc-shaped surface, which enables the connecting piece to rotate in the second direction and be demeshed from the spool.

(14) Preferably, the connecting structure further includes a positioning member and an elastic member; the positioning member includes a connecting plate and a post connected to the connecting plate; the connecting plate is provided with a first connecting part; the post is arranged in the first through hole in a penetrating manner; the substrate of the pawl disc is provided with a second connecting part; the first connecting part is connected to the second connecting part; the elastic member sleeves the post; the connecting piece is provided with a second through hole; the post is arranged in the second through hole in a penetrating manner; the connecting piece is provided with a resisting surface on one side away from the first base plate; and one end of the elastic member resists against the resisting surface, and the other end resists against the connecting plate to reset the connecting piece.
(15) Preferably, the post includes a first end portion and a second end portion opposite to the first end portion; the first end portion is connected to the connecting plate; the post is provided with an annular groove in one end close to the second end portion; a bulge is arranged on the spool; the bulge is arranged in the groove; the second end portion is located on one side of the bulge away from the first end portion; the bulge slides in a circumferential direction of the groove; the post moves in the axial direction to separate the groove from the bulge, and the second end portion resists against one side of the bulge facing the first end portion; the dial is provided with a first positioning part; the connecting plate is provided with a second positioning part on one side away from the post; and the first positioning part is connected to the second positioning part.
(16) Preferably, the spool includes a third through hole penetrating through the first base plate and the second base plate; the post is arranged in the third through hole in a penetrating manner; the spool includes a wall plate; the wall plate is connected to the first base plate and extends towards the second base plate; a hole wall of the third through hole includes the wall plate; the bulge is arranged on one side of the wall plate facing the third through hole; the spool further includes an open slot; the open slot covers the second base plate and extends towards the first base plate; and the open slot is communicated to the second threading hole.
(17) Preferably, the housing further includes a limiting structure; the limiting structure is arranged on one side of the housing away from the dial and resists against the second base plate to limit the spool inside the barrel-shaped main body; the second base plate is provided with a plurality of damping members on one side away from the first base plate; the plurality of damping members are arranged around an edge of the second base plate; the damping members are configured to generate resistance when resisting against the limiting structure; and a force that rotates the spool is greater than the resistance, the spool is rotated relative to the housing.
(18) Preferably, the pawl disc further includes a resistance arm arranged on one side of the substrate away from the dial; the resistance arm has an elasticity to resist against the driving member to provide a resistance; when the dial is rotated in the second direction, and a force applied by the driving member to drive the resistance arm is greater than or equal to the resistance, the pawl disc is rotated in the second direction.
(19) Preferably, the pawl includes an elastic arm connected to the substrate, a second meshing part for being meshed with the first meshing part, and a cylinder connected to the elastic arm and the second meshing part; the cylinder is provided with a connecting hole; a connecting column is arranged on one side of the substrate facing the dial; the connecting hole sleeves the connecting column; the pawl disc further includes a resisting part; the resisting part is arranged on an outer side of the substrate and is adjacent to the connecting column; when the dial is rotated in the first direction, the driving member resists against the resisting part to drive the pawl disc to rotate in the first direction; when the dial is rotated in the second direction, the driving member resists against the second meshing part and moves in a radial direction of the pawl disc; and the cylinder is rotated around the connecting column until the second meshing part is separated from the first meshing part.
(20) Preferably, the dial further includes a first snap; the first snap is arranged on an inner side of

the side wall; the housing is provided with a second snap; the second snap is arranged on an outer side of the barrel-shaped main body; and the first snap is configured to be fastened with the second snap to prevent the dial from being separated from the housing.
(21) Preferably, the lace tightening and loosening device further includes a chassis for placing the lace tightening and loosening device on an external object, the chassis is arranged at one end of the housing away from the dial; the housing further includes a third snap and a fourth snap; the housing is provided with a fifth snap and a sixth snap; and the third snap and the fourth snap are respectively fastened with the fifth snap and the sixth snap.
(22) In a second aspect, the present invention also provides a wearable clothing, which includes a first part, a second part, and a lace, further includes the lace tightening and loosening device described above. The lace tightening and loosening device is configured to tighten or loosen the lace to tighten or loosen the first part and the second part, thereby fixing the wearable clothing onto a human body part; and the wearable clothing includes one of a shoe, a hat, a glove, a medical protective gear, a helmet, a skiing protective gear, and clothes.
(23) In a third aspect, the present invention also provides a storage device, which includes a first part, a second part, and a lace, further including the lace tightening and loosening device described above. The lace tightening and loosening device is configured to tighten or loosen the lace to tighten or loosen the first part and the second part, thereby storing or removing an object into or from the storage device.
(24) In a fourth aspect, the present invention also provides a pawl disc, applied to a lace tightening and loosening device. The pawl disc includes an annular hollow substrate, and a pawl arranged on the substrate; the pawl includes an elastic arm connected to the substrate, and a second meshing part connected to the elastic arm and configured to be meshed with the lace tightening and loosening device; the second meshing part is located at one end of the elastic arm away from the substrate; and the second meshing part is detachably connected to the substrate.
(25) Preferably, the pawl further includes a cylinder arranged between the elastic arm and the second meshing part; a connecting column is arranged on the substrate; a connecting hole is provided in the cylinder; the pawl disc is integrally formed; and the cylinder and the connecting column are detachably connected through the connecting hole.
(26) Compared with the prior art, the pawl disc provided in the present invention is integrally formed, and the cylinder and the connecting column can be detachably connected. After production and manufacturing, during mounting and use, only the cylinder needs to be mounted on the connecting column, which effectively reduces a quantity of parts that achieve a function of the pawl disc, and ensuring the strength of the pawl disc, so that it is convenient for production and manufacturing of the pawl disc, thereby reducing the production costs. The lace tightening and loosening device provided in the present invention can rotate the dial in the first direction to drive the spool to tighten the lace. When the dial is rotated in the second direction to separate the pawl disc and the connecting structure from the spool. The lace can be loosened when an external force is applied to the spool. When no external force is applied to the lace, the spool remains stationary, and when the dial is further rotated in the second direction, the dial, the pawl disc, and the connecting structure remain idling, so that the spool cannot be pushed to rotate in the second direction, thereby ensuring that the lace is not squeezed out, avoiding a situation such as loosening or knotting of the lace inside the spool, and then ensuring the service life of the product. Since the second clamping part with the first clamping part are clamped, the dial is connected to the pawl disc. Therefore, when the dial is lifted, the pawl disc and the connecting structure are also lifted, so that the connecting structure is separated from the spool. This makes the lace drive the spool to rotate in the second direction under the external force, to completely loosen the lace.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to explain the technical solutions of the embodiments of the present invention more clearly, the following will briefly introduce the accompanying drawings used in the embodiments. Apparently, the drawings in the following description are only some embodiments of the present invention. Those of ordinary skill in the art can obtain other drawings based on these drawings without creative work.

(2) FIG. **1** is a three-dimensional diagram of a lace tightening and loosening device according to an embodiment of the present invention;

(3) FIG. **2** is a three-dimensional diagram of a lace tightening and loosening device according to an embodiment of the present invention, viewed in another angle;

(4) FIG. **3** is an exploded view of the lace tightening and loosening device shown in FIG. **1**;

(5) FIG. **4** is an exploded view of the lace tightening and loosening device shown in FIG. **1**, viewed in another angle;

(6) FIG. **5** is a schematic diagram of a housing of the lace tightening and loosening device shown in FIG. **1**;

(7) FIG. **6** is a schematic diagram of the housing shown in FIG. **5**, viewed in another angle;

(8) FIG. **7** is a schematic diagram of a spool of the lace tightening and loosening device shown in FIG. **1**;

(9) FIG. **8** is a schematic diagram of the spool shown in FIG. **7**, viewed in another angle;

(10) FIG. **9** is a partially schematic structural diagram of a housing and a spool of the lace tightening and loosening device shown in FIG. **1**;

(11) FIG. **10** is an enlarged view of part A in FIG. **9**;

(12) FIG. **11** is a schematic diagram of a pawl disc of the lace tightening and loosening device shown in FIG. **1**;

(13) FIG. **12** is a schematic diagram of the pawl disc shown in FIG. **11**, viewed in another angle;

(14) FIG. **13** is a schematic diagram of an initial state of the pawl disc of the lace tightening and loosening device shown in FIG. **11**;

(15) FIG. **14** is a schematic diagram of an initial state of the pawl disc shown in FIG. **13**, viewed in another angle;

(16) FIG. **15** is a schematic diagram of a dial of the lace tightening and loosening device shown in FIG. **1**;

(17) FIG. **16** is a schematic diagram of a connecting structure of the lace tightening and loosening device shown in FIG. **1**;

(18) FIG. **17** is a schematic diagram of a connecting piece of the connecting structure shown in FIG. **16**;

(19) FIG. **18** is a schematic diagram of a rotation direction of a lace tightening and loosening device according to an embodiment of the present invention;

(20) FIG. **19** is a sectional view of a lace tightening and loosening device that rotates in a first direction according to an embodiment of the present invention;

(21) FIG. **20** is a sectional view of a lace tightening and loosening device that rotates in a first direction according to an embodiment of the present invention, viewed in another angle;

(22) FIG. **21** is a sectional view of a lace tightening and loosening device that rotates in a second direction according to an embodiment of the present invention;

(23) FIG. **22** is a sectional view of a lace tightening and loosening device that rotates in a second direction according to an embodiment of the present invention, viewed in another angle;

(24) FIG. **23** is a schematic diagram of a state of completely loosening a lace of a lace tightening and loosening device according to an embodiment of the present invention;

(25) FIG. **24** is a sectional view of a state of completely loosening a lace of a lace tightening and loosening device according to an embodiment of the present invention;

(26) FIG. **25** is a sectional view of a state of tightening a lace of a lace tightening and loosening device according to an embodiment of the present invention;
(27) FIG. **26** is a sectional view of a lace of a lace tightening and loosening device for finely adjusting and loosening a lace according to an embodiment of the present invention;
(28) FIG. **27** is a schematic structural diagram of a shoe according to an embodiment of the present invention;
(29) FIG. **28** is a schematic structural diagram of a hat according to an embodiment of the present invention;
(30) FIG. **29** is a schematic structural diagram of a glove according to an embodiment of the present invention;
(31) FIG. **30** is a schematic structural diagram of a medical protective gear according to an embodiment of the present invention; and
(32) FIG. **31** is a schematic structural diagram of a storage device according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(33) The technical solutions in the embodiments of the present invention will be clearly and completely described below in conjunction with the accompanying drawings in the embodiments of the present invention. Apparently, the described embodiments are only a part of the embodiments of the present invention, rather than all the embodiments. Based on the embodiments in the present invention, all other embodiments obtained by those ordinarily skilled in the art without doing creative work shall fall within the protection scope of the present invention.
(34) To make the aforementioned objectives, features, and advantages of the present invention more comprehensible, specific implementations of the present invention are described in detail below in conjunction with the accompanying drawings.
(35) In order to make a person skilled in the art to better understand the solutions of the present invention, the technical solutions in the embodiments of the present invention are described below with reference to the accompanying drawings in the embodiments of the present invention the described embodiments are merely some rather than all of the embodiments of the present invention. The embodiments in the present invention shall all fall within the protection scope of the present invention.
(36) The terms “first”, “second” and “third” in the specification and claim of the present invention and the attached drawings are used to distinguish different objects rather than to describe a specific sequence. Furthermore, the term “includes”, and any variation thereof, is intended to cover non-exclusive inclusion. For example, a process, method, system, product or device comprising a series of steps or units is not limited to the listed steps or units, but optionally also includes steps or units not listed, or optionally includes other steps or units inherent to those processes, methods, products or devices.
(37) Referring to FIG. **1** and FIG. **2**, an embodiment of the present invention provides a lace tightening and loosening device **1**, configured to tighten and loosen a lace. Specifically, referring to FIG. **3** and FIG. **4**, the lace tightening and loosening device **1** includes a dial **10**, a connecting structure **20**, a pawl disc **30**, a spool **50**, a housing **40**, and a chassis **60**. The dial **10** is connected to the housing **40**. The connecting structure **20**, the pawl disc **30**, and the spool **50** are all located in a space enclosed by the dial **10** and the housing **40**. The chassis **60** is located at one end of the housing **40** away from the dial **10**, to place the lace tightening and loosening device **1** on an object, such as a shoe **2**, a hat **3**, a glove **4**, a medical protective gear **5**, a helmet, a storage device **6**, a skiing protective gear, a bag, and clothes.
(38) Referring to FIG. **5**, the housing **40** includes a barrel-shaped main body **41**, a first meshing part **42** arranged on an inner side of the barrel-shaped main body **41**, a partition plate **43** arranged on the inner side of the barrel-shaped main body **41**, and a first threading hole **40***a* penetrating through the barrel-shaped main body **41**. It can be understood that the barrel-shaped main body **41**

is internally hollow and has openings in two ends. The first meshing part **42** is a shuttle annularly arranged on an inner wall of the barrel-shaped main body **41** and is formed by connecting a plurality of meshing teeth. The partition plate **43** is located at a substantially middle position of the barrel-shaped main body **41**. The first meshing part **42** and the first threading hole **40***a* are respectively located at two end portions of the barrel-shaped main body **41**.
(39) The spool **50** is arranged inside the barrel-shaped main body **41**. Specifically, referring to FIG. **7**, the spool **50** includes a first base plate **51**, a second base plate **52**, and a side wall **53** connecting the first base plate **51** to the second base plate **52**. The first base plate **51** is opposite to the second base plate **52**. The first base plate **51** resists against the partition plate **43**, and the second base plate **52** is located on one side of the first base plate **51** away from the partition plate **43**. The first base plate **51**, the second base plate **52**, and the side wall **53** form a winding region **50***a* for storing a lace. The side wall **53** is provided with a second threading hole **50***b* that penetrates through the side wall **53**. An overall appearance of the spool **50** is "I"-shaped, which is convenient for storing a lace. Specifically, the spool **50** is arranged at one end of the partition plate **43** away from the first meshing part **42**, and the first threading hole **40***a* and the second threading hole **50***b* can be communicated to each other in an overlapping manner to allow the lace to pass through.
(40) The pawl disc **30** is arranged on one side of the first base plate **51** away from the second base plate **52**. Specifically, referring to FIG. **11** and FIG. **12**, the pawl disc **30** includes a substrate **31** with a first through hole **31***a*, a pawl **32** connected to the substrate **31**, and a first clamping part **39** arranged on the substrate **31**. The pawl **32** is configured to be meshed with the first meshing part **42**. Specifically, the pawl disc **30** is located on one side, away from the spool **50**, of the partition plate **43** located inside the barrel-shaped main body **41**, so that the pawl **32** can be meshed with the first meshing part **42**. The pawl **32** can be meshed with or demeshed from the first meshing part **42**. When the pawl **32** is meshed with the first meshing part **42**, the pawl **32** can be separated from the first meshing part **42** in a first direction F. When the pawl **32** is rotated in a second direction R opposite to the first direction F, the first meshing part **42** blocks the rotation of the pawl **32**, thereby preventing the pawl disc **30** from rotating in the second direction R. If the pawl disc **30** needs to be rotated in the second direction R, the pawl **32** needs to be separated from the first meshing part **42**.
(41) The dial **10** is arranged on one side of the pawl disc **30** away from the spool **50**. Specifically, referring to FIG. **15**, the dial **10** includes a cover plate **11**, a side plate **12** connected to one side of the cover plate **11**, a driving member **13** arranged on one side of the cover plate **11** facing the side plate **12**, and a second clamping part **14** arranged on one side of the cover plate **11** facing the side plate **12**. Specifically, the dial **10** is of a roughly cylindrical structure. The cover plate **11** and the side plate **12** are enclosed to form an accommodating cavity **10***a* with an opening, and a portion of the barrel-shaped main body **41** is located in the accommodating cavity **10***a*. In this embodiment, the driving member **13** and the second clamping part **14** are both arranged inside the accommodating cavity **10***a*. It can be understood that the dial **10** is covered at the housing **40**, so that the spool **50** and the pawl disc **30** are both located inside the barrel-shaped main body **41**, thereby achieving a protective effect on the spool **50** and the pawl disc **30**.
(42) Referring to FIG. **3** to FIG. **6**, a limiting structure **48** for resisting connection to limit the spool **50** inside the barrel-shaped main body **41** is arranged on one side of the housing **40** away from the dial **10**. The limiting structure **48** includes a first limiting member **481** and a second limiting member **482**. The first limiting member **481** and the second limiting member **482** resist against one side of the second base plate **52** of the spool **50** away from the first base plate **51**, to limit the spool **50** inside the barrel-shaped main body **41**, thereby preventing the spool **50** from being separated from the housing **40**, and then ensuring that the connecting structure **20**, the pawl disc **30**, and the spool **50** are located between the dial **10** and the housing **40** to achieve linkage of the components. The first limiting member **481** and the second limiting member **482** are opposite to each other on the second base plate **52**. The chassis **60** is located at one end of the housing **40** away from the dial **10**.

(43) In this embodiment, the second clamping part **14** is clamped with the first clamping part **39**. Specifically, the second clamping part **14** is a hook, and the first clamping part **39** is a groove provided in one side of the base plate away from the dial **10**. The hook is hooked with a wall of the groove, thereby connecting the pawl disc **30** to the dial **10** in an axial direction. In this design, when the dial **10** is lifted, the pawl disc **30** can be lifted too through the first clamping part **39** and the second clamping part **14**. When the second clamping part **14** is clamped with the first clamping part **39**, if the dial **10** is rotated, the second clamping part **14** can move inside the first clamping part **39**.
(44) The connecting structure **20** is located inside the barrel-shaped main body **41** and the dial **10**, is arranged in the first through hole **31***a* in a penetrating manner, and connects the pawl disc **30** to the spool **50**. The connecting structure **20** is movably connected to the spool **50**.
(45) The lace tightening and loosening device **1** provided in the present invention has three use functions: tightening a lace, finely adjusting and loosening the lace, and completely loosening the lace. In this embodiment, the dial **10** can be rotated in the first direction F and the second direction R, and can be further lifted in the axial direction. Namely, the dial **10** can move in the axial direction towards one side away from the housing **40**. The first direction F and the second direction R are opposite directions. Referring to FIG. **18**, in this embodiment, using a direction in which the dial **10** faces the housing **40** as an observation angle, using the first direction F as a clockwise direction, and using the second direction R as a counterclockwise direction are taken for example.
(46) Referring to FIG. **19**, when the lace needs to be tightened, the dial **10** is rotated in the first direction F, which drives the driving member **13** to drive the pawl disc **30** to rotate in the first direction F. In this case, the pawl **32** is separated from the first meshing part **42**, which can push the pawl disc **30** to rotate in the first direction F and drive the connecting structure **20** to rotate, thereby driving the spool **50** to rotate in the first direction F to tighten the lace.
(47) Referring to FIG. **21**, when the lace needs to be finely adjusted and loosened, the dial **10** is rotated in the second direction R opposite to the first direction F. The driving member **13** drives the pawl **32** to be separated from the first meshing part **42**, so that the pawl disc **30** can be rotated in the second direction R and drive the connecting structure **20** to rotate. In this case, under an external force, the lace can drive the spool **50** to rotate in the second direction R and be loosened from the spool **50**. When there is no external force, the spool **50** stops rotating, thus achieving fine adjustment on loosening of the lace.
(48) Referring to FIG. **23** and FIG. **24**, when the lace needs to be completely loosened, the dial **10** is lifted in the axial direction under an external force to move away from the housing **40**, and the pawl disc **30** and the connecting structure **20** move away from the spool **50**. In this case, the connecting structure **20** is separated from the spool **50**, and the spool **50** can fspooly rotate inside the housing **40** under the external force, so that the spool **50** can completely loosen the lace under the external force.
(49) Specifically, referring to FIG. **27** to FIG. **30**, a schematic diagram of use of the lace tightening and loosening device **1** to an object is shown. When the lace tightening and loosening device **1** is used, the lace tightening and loosening device **1** is applied to wearable clothing such as a shoe **2**, a hat **3**, a glove **4**, a medical protective gear **5**, a helmet, a skiing protective gear, and clothes. The dial **10** is rotated in the first direction F to tighten the lace, thereby tightening the wearable clothing and fixing the wearable clothing to a human body part such as the head, an arm, a hand, a foot, or the body. However, when the dial **10** is rotated too much, the lace may be strained. In this case, the lace has a tension, which reduces the comfort level of a user. It is necessary to finely adjust and loosen part of the lace. The dial **10** is rotated in the second direction R, so that the connecting structure **20** can be temporarily separated from the spool **50**, to drive, under the tension of the lace, the spool **50** to rotate in the second direction R, thereby loosening the lace. When the lace is adjusted appropriately, the rotation on the dial is stopped, and the spool **50** stops rotating, thereby stopping the loosening of the lace and making the product experience better. There is no need to

completely loosen the lace and then tighten the lace again. When the lace needs to be completely loosened to remove the wearable clothing from the human body part, the dial **10** is lifted, and the lace is pulled to be completely loosened, so that the wearable clothing is loosened and separated from the human body part.

(50) Referring to FIG. **31**, a schematic diagram of use of the lace tightening and loosening device **1** in a storage device is shown. When the lace tightening and loosening device **1** is used, the lace tightening and loosening device **1** is applied to a storage device **6** such as a bag, a backpack, and a storage box. The dial **10** is rotated in the first direction F to tighten the lace, thereby tightening an opening of the storage device **6** and storing an object inside the storage device **6**. However, when the dial **10** is rotated too much, the lace may be strained. In this case, the lace has a tension, which makes the storage device **6** deform or the lace excessively strained. It is necessary to finely adjust and loosen part of the lace. The dial **10** is rotated in the second direction R, so that the connecting structure **20** can be temporarily separated from the spool **50**, to drive, under the tension of the lace, the spool **50** to rotate in the second direction R, thereby loosening the lace. When the lace is adjusted appropriately, the rotation on the dial is stopped, and the spool **50** stops rotating, thereby stopping the loosening of the lace and causing the storage device **6** to have proper tightness during opening tightening. There is no need to completely loosen the lace and then tighten the lace again. When the storage device **6** needs to be opened to take out the object, the dial **10** is lifted, and the lace is pulled to be completely loosened, so that the storage device **6** is completely opened, and the object can be taken out of the storage device **6**.

(51) In this embodiment, two first threading holes **40***a* and two second threading holes **50***b* are included and are in one-to-one correspondence. Two ends of the lace respectively enter the second threading holes **50***b* through the first threading holes **40***a* and are knotted inside the spool **50**, so that the lace can be fixed in the second threading holes **50***b*. Therefore, when the dial **10** is rotated in the first direction F to push the spool **50** to rotate, the lace can be tightened in the winding region **50***a*.

(52) Referring to FIG. **16** and FIG. **17**, the connecting structure **20** includes a positioning member **21**, an elastic member **22**, and a connecting piece **23**. The positioning member **21** includes a connecting plate **211** and a post **212** connected to the connecting plate **211**, and the elastic member **22** sleeves the post **212**. The connecting piece **23** is provided with a second through hole **231***a*. The post **212** and the elastic member **22** are arranged in the second through hole **231***a* in a penetrating manner. The positioning member **21**, the elastic member **22**, and the connecting piece **23** are all arranged in the first through hole **31***a* in a penetrating manner.

(53) In this embodiment, the pawl **32** includes an elastic arm **322** connected to the substrate **31**, a second meshing part **321** for being meshed with the first meshing part **42**, and a cylinder **323** connected to the elastic arm **322** and the second meshing part **321**. The cylinder **323** is provided with a connecting hole **324**. A connecting column **38** is arranged on one side of the substrate **31** facing the dial **10**. The connecting hole **324** sleeves the connecting column **38**. The pawl disc **30** further includes a resisting part **33**. The resisting part **33** is arranged on an outer side of the substrate **31** and is adjacent to the connecting column **38**. When the dial **10** is rotated in the first direction F, the driving member **13** resists against the resisting part **33** to drive the pawl disc **30** to rotate in the first direction F. When the dial **10** is rotated in the second direction R, the driving member **13** resists against the second meshing part **321** and moves in a radial direction of the pawl disc **30**. The cylinder **323** rotates around the connecting column **38** until the second meshing part **321** is separated from the first meshing part **42**.

(54) Specifically, the second clamping part **14** on the dial **10** uses a hook, and the corresponding first clamping part **39** on the pawl disc **30** uses a groove. A perimeter of the groove in a circumferential direction of the substrate **31** is greater than a perimeter of the hook inside the groove, so that the hook can move inside the groove in a length direction of the groove. In this way, when the dial **10** is rotated in the second direction R, the dial **10** and the pawl disc **30** can rotate

relatively by an angle. Specifically, referring to FIG. **20**, when the dial **10** is not rotated, the second clamping part **14** resists against one end surface of the first clamping part **39**. This end surface is defined as a first end surface **391**. At the beginning, the pawl **32** of the pawl disc **30** resists against the first meshing part **42** of the housing **40**. When the dial **10** is rotated in the second direction R, the second clamping part **14** moves along the first clamping part **39** to another end surface of the first clamping part **39**. This end surface is defined as a second end surface **392**. In this case, the dial **10** is rotated relative to the pawl disc **30**. When the dial **10** continues to be rotated in the second direction R, the second clamping part **14** resists against the second end surface **392** of the first clamping part **39**, as shown in FIG. **22**. In this case, the driving member **13** on the dial **10** presses the pawl **32** inwards in the radial direction of the pawl disc **30**, so that the pawl **32** deforms towards a circle center and is separated from the first meshing part **42** of the housing **40**, so that the dial **10** pushes the connecting structure **20** and the pawl disc **30** to rotate in the second direction R. In this case, the lace can drive the spool **50** to rotate in the second direction R under an external force and be loosened from the spool **50**. When the dial **10** remains stationary or rotates without an external force, the pawl **32** resists against the driving member **13**, so that the second clamping part **14** resists against the first end surface **391** of the first clamping part **39**. In this case, when a user touches the dial **10**, the user can feel the stability between the dial **10** and the pawl disc **30**, without any sense of loosening caused by a gap.

(55) To further prevent the pawl disc **30** from rotating in the second direction R when the dial **10** is accidentally touched, causing the spool **50** to rotate in the second direction R to loosen the lace, the pawl disc **30** further includes a resistance arm **37** arranged on one side of the substrate **31** away from the dial **10**. The resistance arm **37** has an elasticity and is configured to resist against the driving member **13** to provide a resistance. When the dial **10** is rotated in the second direction R, a force applied by the driving member **13** to drive the resistance arm **37** is greater than or equal to the resistance, the pawl disc **30** is rotated in the second direction R. By the above structural arrangement, when the dial **10** is touched and rotated in the second direction R, the resistance arm **37** resists against the driving member **13**, so that the rotation of the driving member **13** is affected by the resistance of the resistance arm **37**. When a force applied to the dial **10** is less than the resistance of the resistance arm **37** to the driving member **13**, the driving member **13** and the dial **10** remain stationary, thereby avoiding the impact on the pawl disc **30**, ensuring that the pawl disc **30** and the spool **50** do not rotate, and avoiding occurrence of lace loosening due to accidental touch. In order to avoid product malfunction caused by the dial **10** being separated from the housing **40** in the process of lifting the dial **10** to completely loosen the lace, the dial **10** further includes a first snap **17**. The first snap **17** is arranged on an inner side of the side wall **53**. The housing **40** is provided with a second snap **45**. The second snap **45** is arranged on an outer side of the barrel-shaped main body **41**. The first snap **17** is configured to be fastened with the second snap **45** to prevent the dial **10** from being separated from the housing **40**. Specifically, the first snap **17** is a snap, and the second snap **45** is arranged on an outer edge of the barrel-shaped main body **41** of the housing **40**, so that the first snap **17** can be fastened onto the second snap **45**, thus ensuring that the dial **10** is not separated from the housing **40** when lifted.

(56) The connecting piece **23** includes a main body part **231**, a connecting block **232** arranged on the main body part **231**, and a first tooth **233** arranged at one end of the main body part **231**. The pawl disc **30** further includes a connecting slot **35** extending in an axial direction of the substrate **31**. The connecting slot **35** is communicated to the first through hole **31**. The connecting block **232** is arranged in the connecting slot **35** and can slide along the connecting slot **35**, so that the connecting piece **23** can move in an axial direction of the first through hole **31**. A second tooth **56** is arranged on one side of the first base plate **51** away from the second base plate **52**. The second tooth **56** is configured to be meshed with the first tooth **233**, so that the connecting piece **23** can push the spool **50** to rotate in the first direction F, and the connecting piece **23** can rotate in the second direction R to be demeshed from the spool **50**. Specifically, the first tooth **233** and the

second tooth **56** are both annular shuttles formed by connecting a plurality of meshing teeth. When the connecting piece **23** is rotated in the second direction R and demeshed from the spool **50**, the connecting piece **23** moves along the connecting slot **35**, namely, moves in the axial direction of the first through hole **31***a*. It can be understood that the connecting block **232** is located in the connecting slot **35**, so that the connecting piece **23** can only move in the axial direction of the first through hole **31***a*, and cannot move in a circumferential direction of the first through hole **31***a*. Therefore, when the pawl disc **30** is rotated, under the action of the connecting slot **35** and the connecting block **232**, the connecting piece **23** can rotate with the pawl disc **30**.

(57) In order to reduce the production costs and ensure the strength of the pawl disc **30**, the pawl disc **30** is integrally formed, and the cylinder **323** and the connecting column **38** are detachably connected through the connecting hole **324**. Referring to FIG. **13** and FIG. **14**, an initial state of the pawl disc **30** is shown. After the pawl disc **30** is produced and manufactured, the cylinder **323** is not connected to connecting column **38**. In this case, the connecting hole **324** is separated from the connecting column **38**, and the elastic arm **322** has an elasticity and can have a deformation. When the pawl disc **30** needs to be mounted on the lace tightening and loosening device **1**, before assembling, the cylinder **323** is mounted onto the connecting column **38**, namely, the connecting hole **324** sleeves the connecting column **38**, as shown in FIG. **11** and FIG. **12**. In this design, a quantity of parts of the pawl disc **30** is effectively reduced, the production costs are reduced, and the assembling efficiency is improved.

(58) In order to ensure the meshing strength of the first tooth **233** and the second tooth **56** and facilitate their separation, the first tooth **233** includes a first inclined surface **233***a* and a first arc-shaped surface **233***b* facing away from the first inclined surface **233***a*. The second tooth **56** includes a second inclined surface **561** and a second arc-shaped surface **562** facing away from the second inclined surface **561**. The first inclined surface **233***a* is configured to resist against the second inclined surface **561**, which enables the connecting piece **23** to push the spool **50** to rotate in the first direction F. The first arc-shaped surface **233***b* is configured to resist against the second arc-shaped surface **562**, which enables the connecting piece **23** to rotate in the second direction R and be demeshed from the spool **50**.

(59) Referring to FIG. **25**, when the connecting piece **23** is rotated in the first direction F, the first inclined surface **233***a* resists against the second inclined surface **561**. Since the first inclined surfaces and the second inclined surfaces are both inclined surfaces, they can achieve a tight contact, thereby avoiding separation of the first tooth **233** from the second tooth **56** under a force. This enables the connecting piece **23** to push the spool **50** to rotate in the first direction F, to tighten the lace. Referring to FIG. **26**, when the connecting piece **23** is rotated in the second direction R, the first arc-shaped surface **233***b* and the second arc-shaped surface **562** resist against each other. Since the first arc-shaped surface and the second arc-shaped surface are both arc-shaped surfaces, resistance during their separation can be lower, and a movement distance of the connecting piece **23** in the axial direction of the first through hole **31***a* becomes shorter, facilitating their separation. This enables the spool **50** to loosen a portion of the lace under the tension of the lace. In other embodiments, the second arc-shaped surface **562** and the first arc-shaped surface **233***b* may also be inclined surfaces.

(60) In this embodiment, the connecting plate **211** is provided with a first connecting part **213**. The substrate **31** of the pawl disc **30** is provided with a second connecting part **36**. The first connecting part **213** is connected to the second connecting part **36**. Specifically, one of the first connecting part **213** and the second connecting part **36** is a convex block, and the other one is a groove. The convex block is clamped with the groove to achieve their connection. In detail, the first connecting part **213** is a convex block, and the second connecting part **36** is a groove. There are three first connecting parts **213** and three second connecting parts **36**. The three first connecting parts **213** are arranged around an outer side of the connecting plate **211** and are uniformly distributed, thereby facilitating manufacturing and assembling, and improving the production efficiency. Due to the connection

between the first connecting parts **213** and the second connecting parts **36**, when the dial **10** drives the pawl disc **30** to rotate, the pawl disc **30** drives the positioning member **21** to rotate.

(61) The connecting piece **23** is provided with a resisting surface **234** on one side away from the first base plate **51**. One end of the elastic member **22** resists against the resisting surface **234**, and the other end resists against the connecting plate **211**, to reset the connecting piece **23**. Continuing to refer to FIG. **26**, when the connecting piece **23** is rotated in the second direction R, the first tooth **233** is demeshed from the second tooth **56** of the spool **50**. During the demeshing, the connecting piece **23** moves in the axial direction of the first through hole **31***a*, namely, the connecting piece **23** is away from the spool **50**, thereby pressing the elastic member **22**, causing the elastic member **22** to be compressed and generate a rebound force. When the first tooth **233** and the second tooth **56** are completely separated, the rebound force of the elastic member **22** pushes the connecting piece **23** to move in the axial direction of the first through hole **31***a*, namely, pushing the connecting piece **23** towards the spool **50**, causing the first tooth **233** and the second tooth **56** to be meshed again. Thus, when the dial **10** is rotated in the second direction R, the spool **50** loosens the lace under the tension of the lace. The connecting piece **23** is continuously meshed with and demeshed from the second meshing part **321** of the spool **50** under the rebound force of the elastic member **22**, so that the lace can only be slowly loosened. Namely, the lace is loosened along the meshing teeth of the second meshing part **321** one by one, which achieves fine adjustment of the loosening of the lace.

(62) In this embodiment, the post **212** includes a first end portion **217** and a second end portion **218** opposite to the first end portion **217**. The first end portion **217** is connected to the connecting plate **211**. The post **212** is provided with an annular groove **216** in one end close to the second end portion **218**. A bulge **57** is arranged on the spool **50**. The bulge **57** is arranged in the annular groove **216**. The second end portion **218** is located on one side of the bulge **57** away from the first end portion **217**. The bulge **57** can slide in a circumferential direction of the annular groove **216**. the post **212** moves in the axial direction to separate the annular groove **216** from the bulge **57**, and the second end portion **218** resists against one side of the bulge **57** facing the first end portion **217**. Specifically, a wall of the annular groove **216** and a surface of the bulge **57** are both arc-shaped, so that a frictional force during the relative rotation and movement between the bulge **57** and the annular groove **216** is reduced, wear of the product is reduced, and the service life of the product is prolonged. When the dial **10** is lifted, the dial **10** drives the pawl disc **30** to lift the positioning member **21** together, thereby driving the elastic member **22** and the connecting piece **23** to be lifted together. Thus, the annular groove **216** of the positioning member **21** is separated from the bulge **57** of the spool **50**, and the second end portion **218** of the positioning member **21** to resist against the bulge **57** towards one side of the dial **10**, so that the dial **10**, a connecting mechanism, and the pawl disc **30** are jointly separated from the spool **50**, and the lace inside the spool **50** can be completely loosened.

(63) Referring to FIG. **6** to FIG. **10**, a damping member **55** is arranged on a side surface of the second base plate **52** of the spool **50** away from the first base plate **51**. A plurality of damping members **55** are provided, which are annularly arranged along an edge of the second base plate **52**. The damping member **55** has a third inclined surface **551**, a fourth inclined surface **552**, and a sliding part **553**. One end of the third inclined surface **551** and one end of the fourth inclined surface **552** are respectively connected to the second base plate **52**, and the other ends are respectively connected to two opposite sides of the sliding part **553**. A height of the sliding part **553** protrudes out of a surface of the second base plate **52**, so that the third inclined surface **551** and the fourth inclined surface **552** are inclined relative to the surface of the second base plate **52**. The third inclined surface **551** and the fourth inclined surface **552** can be planes or curved surfaces.

(64) When the spool **50** is rotated, at least one damping member **55** resists against the first limiting member **481** and/or the second limiting member **482** on the housing **40**, thereby generating a resistance to prevent the spool **50** from rotating. Specifically, one of the third inclined surface **551**

or the fourth inclined surface **552** of the damping member **55** resists against a side wall of the first limiting member **481** and/or the second limiting member **482**, thereby generating a resistance. When a force that rotates the spool **50** is greater than the resistance, the first limiting member **481** and/or the second limiting member **482** slides along one of the third inclined surface **551** and the fourth inclined surface **552**, and slides off from the other one. In this process, the first limiting member **481** and/or the second limiting member **482** resists against the sliding part **553** towards a side surface of the spool **50** and slide relative to the sliding part **553**. In this case, the spool **50** is rotated relative to the housing **40**, so that the lace is tightened or loosened. In this embodiment, a maximum perimeter of the second base plate **52** between every two damping members **55** is greater than perimeters of portions, resisting against the second base plate **52**, of the first limiting member **481** and the second limiting member **482**, so that the first limiting member **481** or the second limiting member **482** can be located between the two damping members **55** to achieve resistive sliding or separation from the damping members **55**, thereby controlling a length of tightening or loosening of the lace. When the first limiting member **481** and/or the second limiting member **482** resists against the sliding part **553** and slides relative to the sliding part, the spool **50** moves towards the dial **10**.

(65) In this way, when the dial **10** is rotated in the first direction F, the connecting piece **23** pushes the spool **50** to rotate in the first direction F to tighten the lace. In this case, a pushing force is greater than the resistance generated by the first limiting member **481** and/or the second limiting member **482** to the damping members **55**, and the spool **50** is rotated in the first direction F relative to the housing **40**. When the dial **10** is rotated in the second direction R, the connecting piece **23** is rotated in the second direction R and separated from the spool **50**. In this case, under the tension of the lace, the spool **50** needs to rotate in the second direction R to loosen the lace. When the tension of the lace is greater than the resistance of the first limiting member **481** and/or the second limiting member **482** to the damping members **55**, the spool **50** is rotated in the second direction R relative to the housing **40**. In this case, the lace is slowly loosened under an external tension.

(66) The dial **10** is provided with a first positioning part **16**. A second positioning part **214** is arranged on one side of the connecting plate **211** away from the post **212**. The first positioning part **16** is connected to the second positioning part **214**. Specifically, the first positioning part **16** is a positioning protrusion, and the second positioning part **214** is a groove, so that when the positioning member **21** is assembled, the assembling efficiency can be improved, and damage to a product component due to an assembling error can be avoided. A convex strip **215** is further arranged on one side surface of the connecting plate **211** away from the post **212**, to resist against the cover plate **11** of the dial **10**, so as to reduce a frictional force between the cover plate **11** and the connecting plate **211** and reduce wear.

(67) Referring to FIG. **7** and FIG. **8**, the spool **50** includes a third through hole **50***c* that penetrates through the first base plate **51** and the second base plate **52**, and the post **212** is arranged in the third through hole **50***c* in a penetrating manner. The spool **50** includes a wall plate **54**. The wall plate **54** is connected to the first base plate **51** and extends towards the second base plate **52**. A wall of the third through hole **50***c* includes the wall plate **54**, and the bulge **57** is arranged on one side of the wall plate **54** facing the third through hole **50***c*. In this embodiment, two wall plates **54** are provided, which are opposite to each other. Two bulges **57** are provided. Since the post **212** moves in the axial direction of the third through hole **50***c*, the bulge **57** is separated from the annular groove **216** and resists against the second end portion **218** of the post **212**, so that the wall plates **54** deform. Namely, the two wall plates **54** can deform in opposite directions, so that the bulges **57** can be separated from the annular groove **216**. Therefore, in this embodiment of the present invention, an avoidance space **50***e* is provided in one side of each wall plate **54** away from each bulge **57** to facilitate the deformation of each wall plate **54**.

(68) In this embodiment, in order to allow the lace to be observed during knotting, the spool **50** further includes an open slot **50***d*. The open slot **50***d* is covered at the second base plate **52** and

extends towards the first base plate **51**. The open slot **50***d* is communicated to the second threading hole **50***b*. In this way, the lace passes through the second threading hole **50***b*. During knotting in the open slot **50***d*, a knotting situation and a knot that can accommodate the lace can be directly observed, so that the overall appearance of the product is more beautiful and tidier.

(69) Continuing to refer to FIG. **3**, FIG. **4**, and FIG. **6**, the housing **40** further includes a third snap **46** and a fourth snap **47**. The chassis **60** is provided with a fifth snap **61** and a sixth snap **62**. The third snap **46** and the fourth snap **47** are respectively fastened with the fifth snap **61** and the sixth snap **62**, thereby detachably connecting the housing **40** to the chassis **60**, and further connecting the lace tightening and loosening device **1** to an object through the chassis **60**.

(70) Compared with the prior art, the pawl disc **30** provided in the present invention is integrally formed, and the cylinder **323** and the connecting column **38** can be detachably connected. After production and manufacturing, during mounting and use, only the cylinder **323** needs to be mounted on the connecting column **38**, which effectively reduces a quantity of parts that achieve a function of the pawl disc **30**, and ensuring the strength of the pawl disc **30**, so that it is convenient for production and manufacturing of the pawl disc **30**, thereby reducing the production costs. The lace tightening and loosening device **1** provided in the present invention can rotate the dial **10** in the first direction F to drive the spool **50** to tighten the lace. When the dial **10** is rotated in the second direction R to separate the pawl disc **30** and the connecting structure **20** from the spool **50**. The lace can be loosened when an external force is applied to the spool **50**. When no external force is applied to the lace, the spool **50** remains stationary, and when the dial **10** is further rotated in the second direction R, the dial **10**, the pawl disc **30**, and the connecting structure **20** remain idling, so that the spool **50** cannot be pushed to rotate in the second direction R, thereby ensuring that the lace is not squeezed out, avoiding a situation such as loosening or knotting of the lace inside the spool **50**, and then ensuring the service life of the product. Since the second clamping part **14** with the first clamping part **39** are clamped, the dial **10** is connected to the pawl disc **30**. Therefore, when the dial **10** is lifted, the pawl disc **30** and the connecting structure **20** are also lifted, so that the connecting structure **20** is separated from the spool **50**. This makes the lace drive the spool **50** to rotate in the second direction R under the external force, to completely loosen the lace.

(71) In a second aspect, the present invention further provides wearable clothing. The wearable clothing includes a first part, a second part, and a lace. A gap is reserved between the first part and the second part. The wearable clothing further includes the foregoing lace tightening and loosening device **1**. The lace tightening and loosening device **1** is configured to tighten or loosen the lace to tighten or loosen the first part and the second part, thereby fixing the wearable clothing onto a human body part. The wearable clothing may be a shoe **2**, a hat **3**, a glove **4**, a medical protective gear **5**, a helmet, a skiing protective gear, clothes, and another object.

(72) Referring to FIG. **27**, an embodiment of the present invention further provides a shoe **2**. The shoe **2** includes a shoe body **2***a*, a shoelace **2***b*, and the lace tightening and loosening device **1** of any of the above embodiments. The shoelace **2***b* is the lace, and the lace tightening and loosening device **1** is configured to tighten or loosen the shoelace **2***b*. The lace tightening and loosening device **1** can be fixed on the shoe body **2***a*. In this embodiment, one side of the shoe body **2***a* is the first part, and the other side is the second part. The lace is the shoelace **2***b*. The shoelace **2***b* passes through the two parts to tighten or release the two parts of the shoe body **2***a*, making it easier for a user to put on the shoe.

(73) In this embodiment, using the lace tightening and loosening device **1** of any of the above embodiments is used to tighten or loosen the shoelace **2***b* is simple. A quantity of integrated elements of the lace tightening and loosening device **1** is small, thus reducing the assembling costs, improving the assembling efficiency, making the entire structure compact, and facilitating the miniaturization and weight reduction of the lace tightening and loosening device **1**. Moreover, the small quantity of the elements can also improve the stability of the entire structure, so that the reliability of the lace tightening and loosening device **1** is improved, and the service life of the lace

tightening and loosening device **1** is prolonged.

(74) Referring to FIG. **28**, an embodiment of the present invention further provides a hat **3**. The hat **3** includes a hat body **3***a*, a hat lace **3***b*, and the lace tightening and loosening device **1** of any of the above embodiments. The hat lace **3***b* is configured to tighten or loosen the hat body **3***a*. The hat lace **3***b* is the lace, and the lace tightening and loosening device **1** is configured to tighten or loosen the hat lace **3***b*. The lace tightening and loosening device **1** can be fixed on the hat body **3***a*. In this embodiment, two sides of the hat body **3***a* are respectively the first part and the second part. The lace is the hat lace **3***b*, which connects the two parts to tighten or loosen the hat body **3***a*.

(75) In this embodiment, using the lace tightening and loosening device **1** of any of the above embodiments is used to tighten or loosen the hat lace **3***b* is simple. A quantity of integrated elements of the lace tightening and loosening device **1** is small, thus reducing the assembling costs, improving the assembling efficiency, making the entire structure compact, and facilitating the miniaturization and weight reduction of the lace tightening and loosening device **1**. Moreover, the small quantity of the elements can also improve the stability of the entire structure, so that the reliability of the lace tightening and loosening device **1** is improved, and the service life of the lace tightening and loosening device **1** is prolonged.

(76) Referring to FIG. **29**, an embodiment of the present invention further provides a glove **4**. The glove **4** includes a glove body **4***a*, a glove lace **4***b*, and the lace tightening and loosening device **1** of any of the above embodiments. The glove lace **4***b* is configured to tighten or loosen the glove body **4***a*. The glove lace **4***b* is the lace, and the lace tightening and loosening device **1** is configured to tighten or loosen the glove lace **4***b*. The lace tightening and loosening device **1** can be fixed on the glove body **4***a*. In this embodiment, the glove body **4***a* has a first part and a second part. The first part and the second part are located at an opening. The lace is the glove lace **4***b*, which twists the two parts to tighten or loosen the glove body **4***a*.

(77) In this embodiment, using the lace tightening and loosening device **1** of any of the above embodiments is used to tighten or loosen the glove lace **4***b* is simple. A quantity of integrated elements of the lace tightening and loosening device **1** is small, thus reducing the costs, improving the assembling efficiency, making the entire structure compact, and facilitating the miniaturization and weight reduction of the lace tightening and loosening device **1**. Moreover, the small quantity of the elements can also improve the stability of the entire structure, so that the reliability of the lace tightening and loosening device **1** is improved, and the service life of the lace tightening and loosening device **1** is prolonged.

(78) Referring to FIG. **30**, an embodiment of the present invention further provides a medical protective gear **5**. The medical protective gear **5** includes a medical protective gear body **5***a*, a medical protective gear lace **5***b*, and the lace tightening and loosening device **1** of any of the above embodiments. The medical protective gear lace **5***b* is configured to tighten or loosen the medical protective gear body **5***a*. The medical protective gear lace **5***b* is the lace, and the lace tightening and loosening device **1** is configured to tighten or loosen the medical protective gear lace **5***b*. It can be understood that the medical protective gear **5** can be an arm protective gear (such as a sleeve), a leg protective gear (such as a knee pad and a wrist band), or can be a waist or neck protective gear, which is not limited to the above embodiments. In this embodiment, the medical protective gear body **5***a* is divided into a first part and a second part by an opening of the medical protective gear body **5***a*. The lace is the medical protective gear lace **5***b*, which is wrapped around the opening to tighten or loosen the two parts.

(79) In a third aspect, referring to FIG. **31**, an embodiment of the present application further provides a storage device **6**. The storage device **6** can be, but is not limited to, a backpack, a bag, a storage box, and the like, which includes a storage body **6***a*, a storage belt **6***b*, and the lace tightening and loosening device **1** of any of the above embodiments. The storage belt **6***b* is configured to tighten or loosen the storage body **6***a*. The storage belt **6***b* is the lace, and the lace tightening and loosening device **1** is configured to tighten or loosen the storage belt **6***b*. The lace

tightening and loosening device **1** can be fixed on the storage body **6***a*. In this embodiment, the storage body **6***a* has a first part and a second part. A gap is reserved between the first part and the second part, or the first part and the second part can be separated. The lace is the storage belt **6***b*, which passes through the first part and the second part to tighten or loosen the first part and the second part, thereby storing an object into the storage device **6** or taking out an object from the storage device **6**.
(80) In this embodiment, using the lace tightening and loosening device **1** of any of the above embodiments is used to tighten or loosen the storage belt **6***b* is simple. A quantity of integrated elements of the lace tightening and loosening device **1** is small, thus reducing the assembling costs, improving the assembling efficiency, making the entire structure compact, and facilitating the miniaturization and weight reduction of the lace tightening and loosening device **1**. Moreover, the small quantity of the elements can also improve the stability of the entire structure, so that the reliability of the lace tightening and loosening device **1** is improved, and the service life of the lace tightening and loosening device **1** is prolonged.
(81) The above various embodiments are only used to describe the technical solutions of the present invention, and not intended to limit the present invention. Although the present invention has been described in detail with reference to the foregoing embodiments, those ordinarily skilled in the art should understand that they can still modify the technical solutions described in all the foregoing embodiments, or equivalently replace some or all of the technical features, and these modifications or replacements do not depart the essences of the corresponding technical solutions from the spirit and scope of the technical solutions of all the embodiments of the present invention.

Claims

1. A lace tightening and loosening device, comprising: a base, comprising a barrel-shaped main body, a first meshing part arranged on an inner side of the barrel-shaped main body, a partition plate arranged on the inner side of the barrel-shaped main body, and a first threading hole penetrating through the barrel-shaped main body; a spool, arranged inside the barrel-shaped main body and comprising a first base plate, a second base plate, and a side wall connecting the first base plate to the second base plate, wherein the first base plate is opposite to the second base plate; the first base plate resists against the partition plate, and the second base plate is located on one side of the first base plate away from the partition plate; the first base plate, the second base plate, and the side wall form a winding region for storing a lace; the side wall is provided with a second threading hole that penetrates through the side wall; a pawl disc, arranged on one side of the first base plate away from the second base plate and comprising a substrate with a first through hole, a pawl connected to the substrate, and a first clamping part arranged on the substrate, wherein the pawl is configured to be meshed with the first meshing part; a dial, arranged on one side of the pawl disc away from the spool and comprising a cover plate, a side plate connected to one side of the cover plate, a driving member arranged on one side of the cover plate facing the side plate, and a second clamping part arranged on one side of the cover plate facing the side plate, wherein the cover plate and the side plate are enclosed to form an accommodating cavity with an opening; a portion of the barrel-shaped main body is located in the accommodating cavity; the second clamping part is clamped with the first clamping part; and a connecting structure, arranged in the first through hole in a penetrating manner and connecting the pawl disc to the spool, wherein the connecting structure is movably connected to the spool, wherein when the dial is rotated in a first direction, the driving member drives the pawl disc to rotate in the first direction and drive the connecting structure to rotate, thereby driving the spool to rotate in the first direction to tighten a lace; when the dial is rotated in a second direction opposite to the first direction, the driving member drives the pawl to be separated from the first meshing part, so that the pawl disc rotates in the second direction and drives the connection structure to rotate; meanwhile, under an external force, the lace drives the

spool to rotate in the second direction and be loosened from the spool; when no external force is applied, the spool stops rotating; when the dial is lifted axially away from the base under an external force, the pawl disc and the connecting structure are away from the spool, so that the lace is completely loosened from the spool under the external force; wherein the connecting structure comprises a connecting piece arranged in the first through hole; the connecting piece comprises a main body part, a connecting block arranged at the main body part, and a first tooth arranged at one end of the main body part; the pawl disc further comprises a connecting slot extending in an axial direction of the substrate; the connecting slot is communicated to the first through hole; the connecting block is arranged in the connecting slot and slides along the connecting slot; a second tooth is arranged on one side of the first base plate away from the second base plate; and the second tooth is configured to be meshed with the first tooth, so that the connecting piece pushes the spool to rotate in the first direction, and the connecting piece rotates in the second direction to be demeshed from the spool.

2. The lace tightening and loosening device according to claim 1, wherein the first tooth comprises a first inclined surface and a first arc-shaped surface facing away from the first inclined surface; the second tooth comprises a second inclined surface and a second arc-shaped surface facing away from the second inclined surface; the first inclined surface is configured to resist against the second inclined surface, which enables the connecting piece to push the spool to rotate in the first direction; and the first arc-shaped surface is configured to resist against the second arc-shaped surface, which enables the connecting piece to rotate in the second direction and be demeshed from the spool.

3. The lace tightening and loosening device according to claim 1, wherein the connecting structure further comprises a positioning member and an elastic member; the positioning member comprises a connecting plate and a post connected to the connecting plate; the connecting plate is provided with a first connecting part; the post is arranged in the first through hole in a penetrating manner; the substrate of the pawl disc is provided with a second connecting part; the first connecting part is connected to the second connecting part; the elastic member sleeves the post; the connecting piece is provided with a second through hole; the post is arranged in the second through hole in a penetrating manner; the connecting piece is provided with a resisting surface on one side away from the first base plate; and one end of the elastic member resists against the resisting surface, and the other end resists against the connecting plate to reset the connecting piece.

4. The lace tightening and loosening device according to claim 3, wherein the post comprises a first end portion and a second end portion opposite to the first end portion; the first end portion is connected to the connecting plate; the post is provided with an annular groove in one end close to the second end portion; a bulge is arranged on the spool; the bulge is arranged in the groove; the second end portion is located on one side of the bulge away from the first end portion; the bulge slides in a circumferential direction of the groove; the post moves in the axial direction to separate the groove from the bulge, and the second end portion resists against one side of the bulge facing the first end portion; the dial is provided with a first positioning part; the connecting plate is provided with a second positioning part on one side away from the post; and the first positioning part is connected to the second positioning part.

5. The lace tightening and loosening device according to claim 4, wherein the spool comprises a third through hole penetrating through the first base plate and the second base plate; the post is arranged in the third through hole in a penetrating manner; the spool comprises a wall plate; the wall plate is connected to the first base plate and extends towards the second base plate; a hole wall of the third through hole comprises the wall plate; the bulge is arranged on one side of the wall plate facing the third through hole; the spool further comprises an open slot; the open slot covers the second base plate and extends towards the first base plate; and the open slot is communicated to the second threading hole.

6. The lace tightening and loosening device according to claim 1, wherein the base further

comprises a limiting structure; the limiting structure is arranged on one side of the base away from the dial and resists against the second base plate to limit the spool inside the barrel-shaped main body; the second base plate is provided with a plurality of damping members on one side away from the first base plate; the plurality of damping members are arranged around an edge of the second base plate; the damping members are configured to generate resistance when resisting against the limiting structure; and a force that rotates the spool is greater than the resistance, the spool is rotated relative to the base.

7. The lace tightening and loosening device according to claim 1, wherein the pawl disc further comprises a resistance arm arranged on one side of the substrate away from the dial; the resistance arm has an elasticity to resist against the driving member to provide a resistance; when the dial is rotated in the second direction, and a force applied by the driving member to drive the resistance arm is greater than or equal to the resistance, the pawl disc is rotated in the second direction.

8. A lace tightening and loosening device, comprising: a base, comprising a barrel-shaped main body, a first meshing part arranged on an inner side of the barrel-shaped main body, a partition plate arranged on the inner side of the barrel-shaped main body, and a first threading hole penetrating through the barrel-shaped main body; a spool, arranged inside the barrel-shaped main body and comprising a first base plate, a second base plate, and a side wall connecting the first base plate to the second base plate, wherein the first base plate is opposite to the second base plate; the first base plate resists against the partition plate, and the second base plate is located on one side of the first base plate away from the partition plate; the first base plate, the second base plate, and the side wall form a winding region for storing a lace; the side wall is provided with a second threading hole that penetrates through the side wall; a pawl disc, arranged on one side of the first base plate away from the second base plate and comprising a substrate with a first through hole, a pawl connected to the substrate, and a first clamping part arranged on the substrate, wherein the pawl is configured to be meshed with the first meshing part; a dial, arranged on one side of the pawl disc away from the spool and comprising a cover plate, a side plate connected to one side of the cover plate, a driving member arranged on one side of the cover plate facing the side plate, and a second clamping part arranged on one side of the cover plate facing the side plate, wherein the cover plate and the side plate are enclosed to form an accommodating cavity with an opening; a portion of the barrel-shaped main body is located in the accommodating cavity; the second clamping part is clamped with the first clamping part; and a connecting structure, arranged in the first through hole in a penetrating manner and connecting the pawl disc to the spool, wherein the connecting structure is movably connected to the spool, wherein when the dial is rotated in a first direction, the driving member drives the pawl disc to rotate in the first direction and drive the connecting structure to rotate, thereby driving the spool to rotate in the first direction to tighten a lace; when the dial is rotated in a second direction opposite to the first direction, the driving member drives the pawl to be separated from the first meshing part, so that the pawl disc rotates in the second direction and drives the connection structure to rotate; meanwhile, under an external force, the lace drives the spool to rotate in the second direction and be loosened from the spool; when no external force is applied, the spool stops rotating; when the dial is lifted axially away from the base under an external force, the pawl disc and the connecting structure are away from the spool, so that the lace is completely loosened from the spool under the external force; wherein the pawl comprises an elastic arm connected to the substrate, a second meshing part for being meshed with the first meshing part, and a cylinder connected to the elastic arm and the second meshing part; the cylinder is provided with a connecting hole; a connecting column is arranged on one side of the substrate facing the dial; the connecting hole sleeves the connecting column; the pawl disc further comprises a resisting part; the resisting part is arranged on an outer side of the substrate and is adjacent to the connecting column; when the dial is rotated in the first direction, the driving member resists against the resisting part to drive the pawl disc to rotate in the first direction; when the dial is rotated in the second direction, the driving member resists against the second meshing part and moves in a radial

direction of the pawl disc; and the cylinder is rotated around the connecting column until the second meshing part is separated from the first meshing part.
9. A lace tightening and loosening device, comprising: a base, comprising a barrel-shaped main body, a first meshing part arranged on an inner side of the barrel-shaped main body, a partition plate arranged on the inner side of the barrel-shaped main body, and a first threading hole penetrating through the barrel-shaped main body; a spool, arranged inside the barrel-shaped main body and comprising a first base plate, a second base plate, and a side wall connecting the first base plate to the second base plate, wherein the first base plate is opposite to the second base plate; the first base plate resists against the partition plate, and the second base plate is located on one side of the first base plate away from the partition plate; the first base plate, the second base plate, and the side wall form a winding region for storing a lace; the side wall is provided with a second threading hole that penetrates through the side wall; a pawl disc, arranged on one side of the first base plate away from the second base plate and comprising a substrate with a first through hole, a pawl connected to the substrate, and a first clamping part arranged on the substrate, wherein the pawl is configured to be meshed with the first meshing part; a dial, arranged on one side of the pawl disc away from the spool and comprising a cover plate, a side plate connected to one side of the cover plate, a driving member arranged on one side of the cover plate facing the side plate, and a second clamping part arranged on one side of the cover plate facing the side plate, wherein the cover plate and the side plate are enclosed to form an accommodating cavity with an opening; a portion of the barrel-shaped main body is located in the accommodating cavity; the second clamping part is clamped with the first clamping part; and a connecting structure, arranged in the first through hole in a penetrating manner and connecting the pawl disc to the spool, wherein the connecting structure is movably connected to the spool, wherein when the dial is rotated in a first direction, the driving member drives the pawl disc to rotate in the first direction and drive the connecting structure to rotate, thereby driving the spool to rotate in the first direction to tighten a lace; when the dial is rotated in a second direction opposite to the first direction, the driving member drives the pawl to be separated from the first meshing part, so that the pawl disc rotates in the second direction and drives the connection structure to rotate; meanwhile, under an external force, the lace drives the spool to rotate in the second direction and be loosened from the spool; when no external force is applied, the spool stops rotating; when the dial is lifted axially away from the base under an external force, the pawl disc and the connecting structure are away from the spool, so that the lace is completely loosened from the spool under the external force; wherein the dial further comprises a first snap; the first snap is arranged on an inner side of the side wall; the base is provided with a second snap; the second snap is arranged on an outer side of the barrel-shaped main body; and the first snap is configured to be fastened with the second snap to prevent the dial from being separated from the base.
10. The lace tightening and loosening device according to claim 1, further comprising a chassis for placing the lace tightening and loosening device on an external object, wherein the chassis is arranged at one end of the base away from the dial; the base further comprises a third snap and a fourth snap; the base is provided with a fifth snap and a sixth snap; and the third snap and the fourth snap are respectively fastened with the fifth snap and the sixth snap.
11. Wearable clothing, comprising a first part, a second part, and a lace, further comprising the lace tightening and loosening device according to claim 1, wherein the lace tightening and loosening device is configured to tighten or loosen the lace to tighten or loosen the first part and the second part, thereby fixing the wearable clothing onto a human body part; and the wearable clothing comprises one of a shoe, a hat, a glove, a medical protective gear, a helmet, a skiing protective gear, and clothes.
12. A storage device, comprising a first part, a second part, and a lace, further comprising the lace tightening and loosening device according to claim 1, wherein the lace tightening and loosening device is configured to tighten or loosen the lace to tighten or loosen the first part and the second

part, thereby storing or removing an object into or from the storage device.

13. A pawl disc, applied to a lace tightening and loosening device, wherein the pawl disc comprises an annular hollow substrate, and a pawl arranged on the substrate; the pawl comprises an elastic arm connected to the substrate, and a second meshing part connected to the elastic arm and configured to be meshed with the lace tightening and loosening device; the second meshing part is located at one end of the elastic arm away from the substrate; and the second meshing part is detachably connected to the substrate; wherein the pawl further comprises a cylinder arranged between the elastic arm and the second meshing part; a connecting column is arranged on the substrate; a connecting hole is provided in the cylinder; the pawl disc is integrally formed; and the cylinder and the connecting column are detachably connected through the connecting hole.